

Optical and Thermal Imaging in Agriculture: A Systematic Review of AI-Based Sensing Technologies, Crop Applications, and Algorithmic Trends

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Abstract

The integration of artificial intelligence (AI) with optical and thermal imaging technologies in agricultural systems has expanded rapidly in recent years, yet the scientific literature addressing this intersection remains fragmented across sensing modalities, acquisition platforms, and algorithmic approaches. This systematic literature review synthesises evidence on the application of AI and machine learning techniques to optical and thermal image analysis in agricultural and agro-industrial contexts, covering peer-reviewed journal articles published between 2023 and 2025. The search was conducted in ScienceDirect and IEEE Xplore following PRISMA 2020 guidelines, combining the domains of artificial intelligence, imaging technologies — including RGB, multispectral, hyperspectral, and thermal modalities — and agriculture. From an initial pool of 496 records, 91 studies met the inclusion criteria and were retained for analysis. Results indicate that RGB sensors and UAV-based platforms constitute the dominant technological configuration in the reviewed literature, while thermal and multispectral imaging represent growing but still secondary modalities. Convolutional neural networks (CNNs) were the most frequently employed AI architecture, identified in 52 of the 91 included studies either exclusively or in combination with other models, within a broader deep learning family that spans 54 studies. Cereal crops and fruit crops were the most studied agricultural categories, and pest and disease detection emerged as the leading application objective, followed by irrigation and water status monitoring and maturity or phenology assessment. A cross-analysis of crop types and applications revealed a non-uniform distribution of research efforts, with several combinations — including legumes, vineyards, and forestry ecosystems — remaining underexplored. These findings highlight both the consolidation of deep learning approaches in optical and thermal agricultural image analysis and the need for standardised datasets, cross-domain validation strategies, and expanded integration of multimodal sensing technologies to improve the generalisability and practical applicability of AI-based agricultural monitoring systems.

1 Introduction

Global agricultural systems face mounting pressure from population growth, climate variability, and resource constraints, demanding more efficient and scalable approaches to crop production and management [1]. The digital transformation of agriculture has accelerated in response, driven by the convergence of advanced sensing technologies, data analytics platforms, and artificial intelligence (AI) [2]. Modern precision agriculture increasingly relies on data-driven approaches to monitor crop health, optimise resource allocation, and improve productivity across diverse agronomic contexts [3].

Among the enabling technologies underpinning this transformation, optical and thermal imaging systems have emerged as particularly powerful tools for non-invasive, continuous observation of plant physiology, environmental stress conditions, and crop development. This review focuses specifically on the intersection of AI and machine learning with optical imaging modalities — including RGB, multispectral, hyperspectral, and thermal infrared — applied within agricultural and agro-industrial

settings. These modalities collectively span the visible and infrared spectrum, offering complementary windows into biochemical composition, canopy temperature, and structural plant properties that no single sensor type can capture alone [4].

Traditional RGB cameras capture visible-light reflectance and support tasks such as crop classification, weed detection, and phenological analysis. However, their performance degrades substantially under low-illumination conditions, and they are entirely insensitive to phenomena with no visible spectral signature, such as early-stage water stress or subsurface physiological anomalies. Multispectral and hyperspectral sensors extend the observable range to wavelengths closely associated with plant pigmentation, water content, and biochemical composition, enabling the detection of stress conditions prior to visible symptom development [5].

Thermal infrared imaging addresses the limitations of passive optical systems by detecting the electromagnetic radiation emitted by any object above absolute zero. Infrared thermography operates primarily within the long-wave infrared band (8–14 μm), which provides high sensitivity to surface temperature gradients under a wide range of ambient conditions [6]. In agricultural applications, thermal patterns across plant canopies function as indirect physiological indicators: elevated canopy temperatures signal reduced transpiration associated with stomatal closure under water deficit, while localised thermal anomalies can indicate the onset of fungal infection, vascular disease, or pest-induced tissue damage prior to visible symptom expression [7]. These properties make thermography particularly valuable for precision irrigation scheduling, early disease surveillance, and non-destructive crop quality assessment throughout the production cycle.

The operational capacity of imaging systems in agricultural environments has been substantially extended by advances in data acquisition platforms. Unmanned aerial vehicles (UAVs) have become the dominant platform for field-scale image acquisition due to their operational flexibility, sub-decimetres spatial resolution, and relatively low deployment cost compared with manned aircraft or satellite systems [8]. Equipped with multispectral, hyperspectral, and thermal sensors, UAVs enable the rapid acquisition of multi-dimensional data including canopy structure, spectral reflectance, and surface temperature, supporting accurate diagnosis of crop water stress, disease incidence, and nutrient deficiencies. The integration of these platforms with Internet of Things (IoT) infrastructures enables real-time environmental monitoring and supports dynamic, condition-responsive crop management decisions.

The expansion of image-based agricultural monitoring has been closely coupled with advances in artificial intelligence and machine learning. Deep learning architectures, and convolutional neural networks (CNNs) in particular, have fundamentally transformed the capacity to automatically interpret large volumes of agricultural imagery, identifying complex visual patterns associated with crop diseases, pest infestations, abiotic stress, and developmental stages [9]. More recently, transformer-based architectures and hybrid approaches combining CNNs with attention mechanisms have further extended model representational capacity, improving generalisation across heterogeneous field environments and multiple sensing modalities [10].

Despite these advances, the scientific literature on AI-based optical and thermal agricultural image analysis remains fragmented. Studies differ substantially in their choice of sensing modality, acquisition platform, target crop, AI methodology, and performance evaluation protocol, making direct comparison across contributions difficult and systematic identification of dominant technological trends elusive. Persistent barriers include the scarcity of large annotated datasets, limited model generalisation across geographic regions and crop varieties, and the challenge of fusing information from heterogeneous optical and thermal sensor types. No comprehensive synthesis specifically addressing the intersection of RGB, multispectral, hyperspectral, and thermal imaging modalities with AI approaches for the most recent period (2023–2025) has been published to date.

This systematic literature review therefore aims to address that gap. Following the PRISMA 2020 guidelines, we synthesise peer-reviewed evidence from studies applying artificial intelligence and machine learning to optical and thermal agricultural image analysis, characterise the technological configurations, algorithmic approaches, and application domains that define the current state of the art, and identify priorities for future research in AI-driven precision agriculture.

2 Methods

2.1 Search Strategy

The systematic literature search was conducted in accordance with the PRISMA 2020 guidelines. The objective was to identify studies applying artificial intelligence (AI) or machine learning algorithms to the processing of optical and thermal images — including RGB, multispectral, hyperspectral, and thermal infrared modalities — in agricultural or agro-industrial contexts. The search covered the period from 2023 to 2025 and included only peer-reviewed journal articles written in English.

The search query combined three conceptual domains: (1) artificial intelligence or machine learning, (2) optical or thermal imaging technologies, and (3) agriculture or agro-industry. The final search string was defined as follows: (*“artificial intelligence”* OR *“machine learning”*) AND (*“thermal imaging”* OR *“thermography”* OR *“image analysis”* OR *“image processing”*) AND (*“agriculture”* OR *“agroindustry”*).

This combination was designed to ensure the retrieval of studies addressing the intersection between AI algorithms and image-based analysis — encompassing both thermal and broader optical modalities — within agricultural and agro-industrial systems. The inclusion of thermal imaging terms as anchor concepts reflects the particular relevance of this modality for physiological crop monitoring, while the broader imaging terms allowed the capture of RGB, multispectral, and hyperspectral studies that co-occur in this research space.

2.2 Information Sources and Databases

Scientific databases relevant to engineering and agricultural research were selected to ensure adequate coverage of the topic. The search was performed in ScienceDirect and IEEE Xplore, which together provide extensive indexing of research on artificial intelligence, imaging technologies, and precision agriculture engineering. ScienceDirect was prioritised for its broad coverage of applied agricultural and biosystems engineering journals, while IEEE Xplore was selected for its coverage of computational and signal processing literature. The restriction to these two databases constitutes a recognised limitation of the present review: broader searches incorporating Scopus, Web of Science, or domain-specific repositories such as PubMed may yield additional relevant studies, particularly from plant science, food technology, and agronomy journals. This limitation is acknowledged explicitly in the Conclusions section.

All retrieved references were exported to a reference management system (Mendeley) for organisation, screening, and removal of duplicates prior to the selection process. Only peer-reviewed journal articles were considered for inclusion at this stage; conference proceedings, book chapters, and grey literature were excluded.

2.3 Inclusion and Exclusion Criteria

Inclusion and exclusion criteria were established *a priori* to guide the screening and selection of studies.

Inclusion criteria:

- Studies that apply image processing techniques.
- Studies related to agriculture or agro-industrial applications.

Exclusion criteria:

- Studies not addressing the above inclusion criteria.
- Review papers or theoretical studies without experimental or applied image analysis.

These criteria ensured that only primary research articles applying artificial intelligence algorithms to image processing within agricultural contexts were retained for detailed examination.

2.4 Study Selection Process

The study selection process was conducted following the PRISMA 2020 recommendations to ensure transparency, reproducibility, and methodological rigor. After exporting all retrieved records from the selected databases into the reference management software Mendeley, an automated duplicate detection process was performed, followed by a manual verification to ensure the accurate removal of redundant entries.

The screening phase was carried out in two sequential stages. In the first stage, titles and abstracts were independently reviewed by at least two members of the research team to assess their relevance with respect to the predefined inclusion and exclusion criteria. Disagreements between reviewers were resolved through discussion and consensus. This stage aimed to rapidly eliminate studies that were clearly unrelated to optical and thermal agricultural image analysis, artificial intelligence methodologies, or experimental applications. Only studies that explicitly addressed image-based analysis within agricultural or agro-industrial contexts were retained for further evaluation.

In the second stage, full-text articles were retrieved and examined in detail to confirm their eligibility. This assessment focused on verifying the actual implementation of artificial intelligence or machine learning techniques, the use of image data as a primary information source, and the relevance of the agricultural application. Studies that did not provide sufficient methodological detail, lacked experimental validation, or focused exclusively on theoretical or review-based discussions were excluded at this stage. Full-text assessment was likewise conducted with cross-verification between reviewers to ensure consistency of the eligibility decisions.

Any remaining ambiguities encountered during the selection process were resolved through discussion among the research team to maintain methodological consistency. The entire selection workflow was systematically documented to ensure traceability and to support the construction of the PRISMA flow diagram, which summarises the identification, screening, eligibility, and inclusion phases of the review.

2.5 Data Extraction and Coding

A structured and standardized data extraction framework was developed to ensure consistent and comprehensive information retrieval from each included study. The extraction process was designed to capture both contextual and technical dimensions relevant to agricultural image analysis using artificial intelligence.

The extracted bibliographic data included authorship, publication year, and geographical context, enabling temporal and regional trend analysis. From a domain perspective, information regarding crop type, agricultural system (open field or greenhouse), and application objective was recorded. Technical variables encompassed the imaging modality (thermal, RGB, multispectral, or hyperspectral), acquisition platform (drone, satellite, or ground-based systems), spatial and temporal resolution, and preprocessing techniques applied prior to model training.

Regarding artificial intelligence methodologies, each study was coded according to the type of algorithm or model employed, such as convolutional neural networks, traditional machine learning classifiers, or hybrid approaches. Additional variables included the computational task performed (classification, detection, regression, or segmentation), target variable, evaluation metrics, and reported performance outcomes.

To reduce semantic inconsistencies across studies, a normalization process was applied during the coding phase, grouping algorithmic variants under unified categories. Data extraction was performed by a primary reviewer and subsequently verified by a second member of the research team to ensure internal coherence and terminological consistency. This structured coding strategy facilitated downstream synthesis and comparative analysis across heterogeneous studies.

2.6 Quality Assessment

A qualitative methodological assessment was conducted to evaluate the robustness and reliability of the included studies. Although no numerical scoring system was applied, each article was systematically examined against a set of predefined quality indicators derived from best practices in experimental artificial intelligence research.

The assessment criteria included: (i) clarity of research objectives and problem formulation, (ii) adequacy and description of the dataset, including sample size and acquisition conditions, (iii) transparency and reproducibility of the methodological pipeline, including preprocessing, model architecture, and training procedures, and (iv) validity of the evaluation strategy, particularly the use of appropriate performance metrics and validation protocols.

Special attention was given to the reporting of experimental limitations, generalization capacity, and potential sources of bias, such as dataset imbalance or overfitting. Studies that lacked sufficient methodological detail or failed to justify their evaluation approach were considered to have lower analytical reliability and were interpreted cautiously during synthesis.

This quality assessment process ensured that the resulting evidence synthesis was grounded in studies meeting acceptable standards of methodological rigor, thereby strengthening the credibility of the review findings.

2.7 Data Synthesis and Analysis

The synthesis of extracted data was conducted using a qualitative and descriptive analytical approach, given the heterogeneity of agricultural contexts, imaging modalities, and artificial intelligence techniques identified in the literature. Rather than performing a meta-analysis, the review focused on identifying recurring methodological patterns, dominant algorithmic trends, and emerging application areas.

Studies were grouped according to imaging modality, type of artificial intelligence model, and agricultural application. This categorization enabled the identification of dominant approaches, such as the prevalence of convolutional neural networks in image-based classification and detection tasks, as well as the increasing integration of thermal imaging for stress detection and monitoring applications.

Comparative analysis was performed to examine relationships between model selection, sensing platforms, and reported performance outcomes. This process facilitated the identification of methodological strengths, such as improved accuracy through deep learning architectures, and recurring challenges, including data scarcity and limited generalization across crops or environments.

The synthesis also highlighted research gaps and future directions, particularly the need for standardized datasets, cross-domain validation, and multimodal sensor fusion strategies. Overall, this structured synthesis provides a coherent overview of how artificial intelligence has been operationalised in optical and thermal agricultural image analysis, offering both a consolidated state of the art and a foundation for future research.

3 Results

3.1 Study Selection

A total of 496 records were retrieved from two electronic databases: ScienceDirect ($n = 483$) and IEEE Xplore ($n = 13$). After removing one duplicate record, 495 records remained for title and abstract screening. During this phase, 393 records were excluded for not meeting the inclusion criteria related to the use of image processing techniques or agricultural applications, yielding 102 reports for full-text retrieval.

All 102 reports were successfully accessed ($n = 0$ not retrieved). After full-text assessment for eligibility, 11 studies were excluded: 5 for lacking AI applications, 4 for lacking agricultural relevance, and 2 for focusing on animal-related applications. The remaining 91 studies met the inclusion criteria and were included in the final review.

Relevant information from each included study was collected through a systematic data extraction process designed to ensure a homogeneous characterization of both contextual and technical aspects of the analyzed research. This process encompassed variables related to the agricultural domain, type of production system, image acquisition modality, sensing platform, and application scenario, as well as the agronomic objectives addressed. The standardization of these criteria enabled meaningful comparisons across studies conducted in different regions, crops, and experimental conditions, thereby reducing methodological heterogeneity and facilitating a coherent synthesis of the available evidence.

This approach ensured data traceability and supported a structured evaluation of artificial intelligence applications in agricultural image analysis.

In addition, variables specifically related to the applied artificial intelligence approaches were extracted and systematically coded, including model type, computational task, and reported performance metrics. This coding process allowed for the normalization of terminological variations across the literature and enabled a comparative analysis aimed at identifying trends in the use of different algorithmic families. Based on this synthesis, studies were classified according to the categories of models employed, and their relationships with agricultural applications were examined, providing a comprehensive overview of the state of the art. This procedure strengthened the analytical consistency of the review and facilitated the identification of recurring patterns, methodological limitations, and future research opportunities in the application of artificial intelligence to agricultural image analysis.

The complete selection process is illustrated in Figure 1.

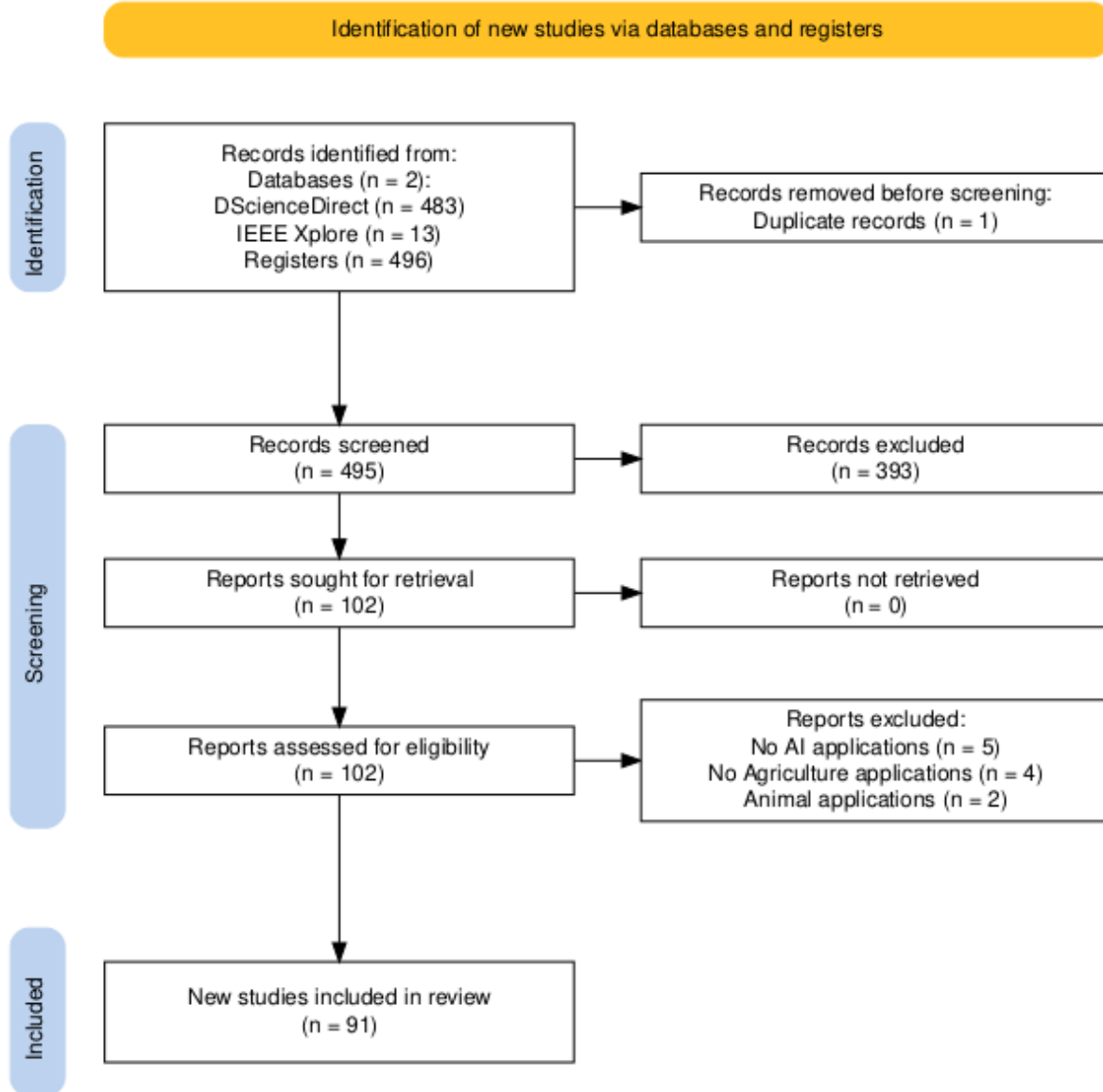


Figure 1: PRISMA flow diagram illustrating the identification, screening, and inclusion process of studies retrieved from ScienceDirect and IEEE Xplore between 2023 and 2025.

3.2 Data Extraction and Analysis

For each included study, the following information was extracted and organized in an Excel spreadsheet: Inclusion number, reference (APA format), year, region/country, type of optical capture sensor, resolution (spatial/temporal), platform (drone, satellite, manual), crop type, scenario (greenhouse or open field), main findings summary, type of AI model or algorithm, performed task (classification, detection, prediction, etc.), analyzed variable, agricultural application (irrigation, pest detection, stress, maturity assessment), performance metrics (accuracy, recall, RMSE, etc.), percentage/value, and reported limitations or future directions.

The dataset was subsequently imported into Python for quantitative analysis. A normalization process was performed on the names of AI models and algorithms to unify variations in terminology across studies. This standardization enabled a frequency analysis and the generation of categories representing the most frequently applied AI approaches in the reviewed literature.

The overall analysis of the technologies employed in the reviewed studies reveals a clear predominance of RGB sensors and UAV/drone platforms as the primary technological configuration. Figure 1 shows that RGB sensors account for the largest number of publications, followed by multisensor approaches that combine different spectral modalities, reflecting a growing trend toward data fusion to enhance the robustness of artificial intelligence models. In terms of acquisition platforms, UAVs clearly dominate due to their operational flexibility and high spatial resolution, while manually collected datasets and fixed platforms represent a significant but smaller proportion. From an agronomic perspective, cereal crops constitute the most studied category, followed by fruit crops and multi-crop studies. Regarding applications, pest and disease detection clearly leads the literature, followed by irrigation/water status monitoring and maturity/phenology analysis, confirming that most AI developments are oriented toward early diagnosis and physiological crop monitoring.

3.3 Sensing Technologies and Acquisition Platforms

The distribution of optical sensor types across the reviewed studies reveals a marked predominance of RGB imagery, which accounts for the largest share of publications (see Figure 2, panel a). This prevalence reflects the widespread availability, low acquisition cost, and straightforward integration of RGB sensors with AI-based image processing pipelines [11]. Multisensor configurations constitute the second most common category, indicating a growing interest in data fusion strategies that combine complementary sensing modalities to improve model robustness and predictive accuracy. Thermal sensors appear in a moderate proportion of studies, primarily in applications related to water stress detection and irrigation management [12, 13]. Hyperspectral and multispectral sensors are comparatively less frequent, likely due to higher hardware costs, greater data complexity, and the specialised preprocessing requirements associated with these modalities [14, 15, 16].

Regarding data acquisition platforms (see Figure 2, panel b), unmanned aerial vehicles (UAVs or drones) represent the most commonly employed system, reflecting their operational flexibility, high spatial resolution, and suitability for field-scale monitoring tasks. Studies relying on manually collected or publicly available datasets constitute the second largest group, frequently used for algorithm development, benchmarking, and transfer learning experiments. Fixed or static platforms — including tower-mounted or greenhouse-integrated systems — account for a smaller but consistent proportion of studies, while ground-based mobile platforms appear in a limited number of publications. The dominance of UAV-based and manual acquisition strategies highlights both the maturity of drone-based sensing in precision agriculture and the continued relevance of curated benchmark datasets for AI model development [17, 18].

3.4 Crop Categories and Agricultural Applications

The analysis of crop categories (see Figure 2, panel c) reveals that cereal crops dominate the reviewed literature, consistent with their global economic importance and the comparative availability of large-scale labelled datasets for training AI models [12, 19, 20, 21]. Fruit crops represent the second most frequently studied category, often in the context of disease detection, yield estimation, and postharvest quality assessment, reflecting the high commercial value and strict visual quality requirements of this

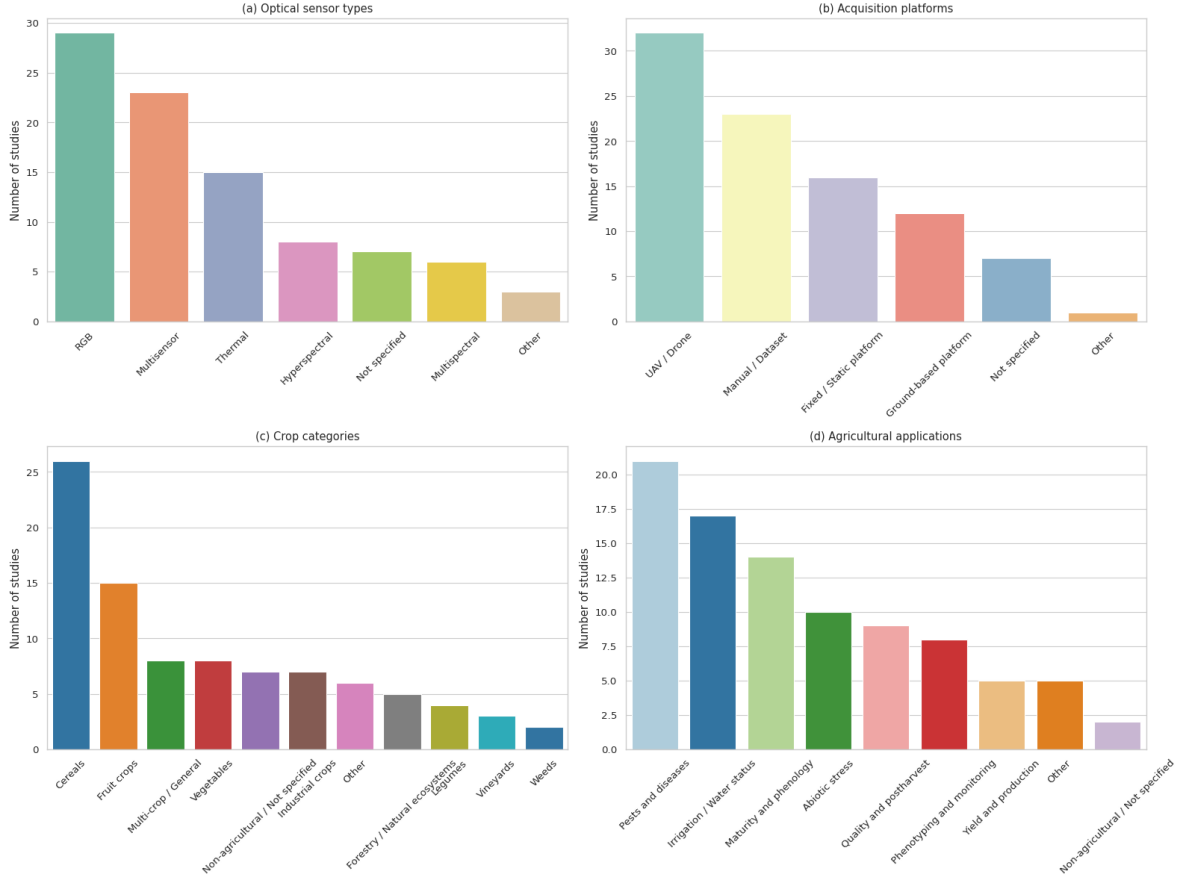


Figure 2: Distribution of the reviewed studies across four main dimensions: optical sensor types, acquisition platforms, crop categories, and agricultural applications. The results show a clear predominance of RGB sensors and UAV/drone platforms, as well as a concentration of research on cereal crops and pest and disease detection applications.

production system [22, 23]. Multi-crop and generalised agricultural studies account for a significant proportion of the literature, suggesting ongoing efforts to develop transferable AI models applicable across diverse crops and environments. Vegetables, industrial crops, and legumes appear with moderate frequency, while vineyards, forestry and natural ecosystems, and weed-focused studies remain comparatively underrepresented, indicating persistent gaps in the coverage of certain agricultural domains [24, 25, 26, 27].

In terms of agricultural applications (see Figure 2, panel d), pest and disease detection is the most frequently addressed objective, underscoring the economic impact of biotic stressors and the recognised value of early detection systems for crop protection [28, 24, 15]. Irrigation and water status monitoring constitutes the second most common application, followed by studies on crop maturity and phenological development. Abiotic stress detection, postharvest quality assessment, and phenotyping and general monitoring are moderately represented, while yield and production estimation appears less frequently, likely owing to the data requirements and modelling complexity associated with reliable predictions. The cross-analysis of crop types and applications (see Figure 3) further reveals an uneven distribution of research attention: cereal crops are broadly studied across irrigation, maturity, and disease-related objectives, fruit crops show a concentration in quality assessment and disease detection, and several crop-application combinations — particularly involving legumes, vineyards, and natural ecosystems — remain largely unexplored, representing clear opportunities for future investigation [29, 25].

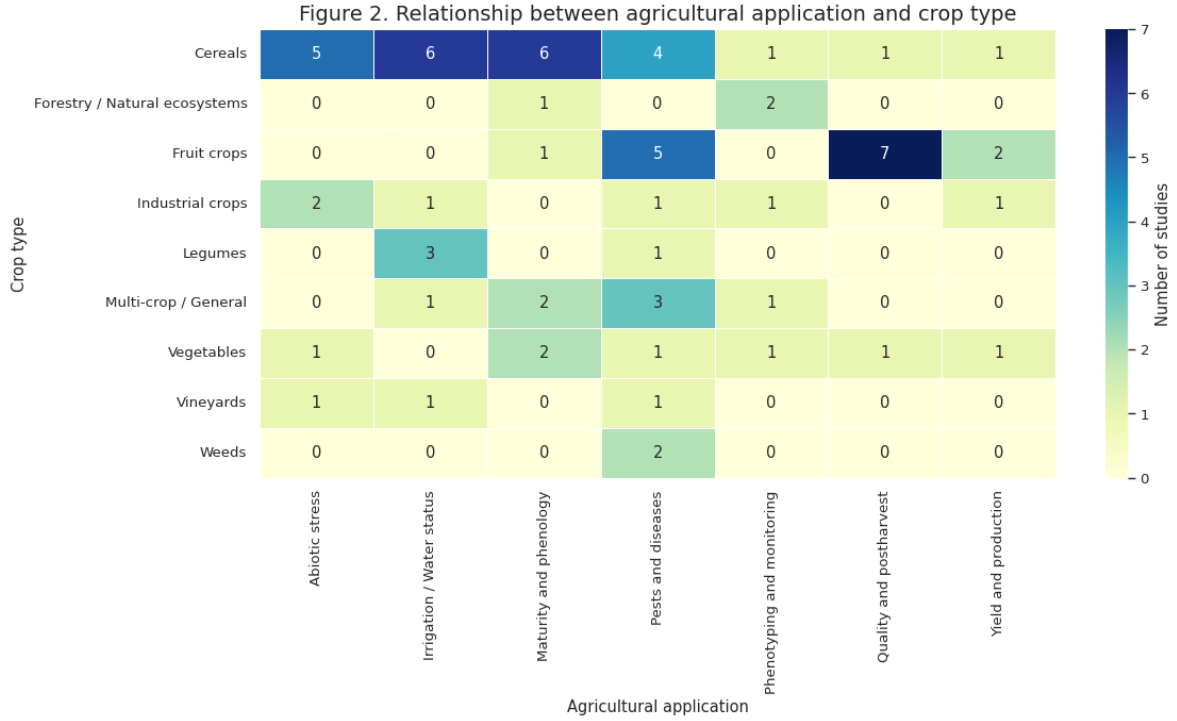


Figure 3: Relationship between crop types and agricultural applications identified in the reviewed studies. The heatmap highlights the frequency of studies for each crop–application combination, revealing research concentrations in specific agricultural domains.

3.5 Artificial Intelligence Models and Methodological Approaches

Table 1 summarises the distribution of AI model families identified across the reviewed studies. Deep learning constitutes the dominant methodological category, with 54 studies employing deep learning architectures, underscoring the widespread adoption of data-driven representation learning in contemporary agricultural research. Tree-based machine learning methods, including Random Forest and decision tree variants, represent the second most common family (16 studies), followed by kernel-based approaches such as Support Vector Machines (11 studies), highlighting the continued relevance of classical machine learning for structured data and lower-complexity classification tasks. Neural network-based methods beyond deep learning (9 studies), regression models (9 studies), and boosting algorithms (7 studies) account for a moderate share of the literature. Transformer-based deep learning architectures (5 studies), while less frequent, represent an emerging trend, suggesting a gradual shift toward models capable of capturing long-range spatial and spectral dependencies in agricultural imagery.

At the level of individual models (Table 2), convolutional neural networks (CNNs) are by far the most frequently reported architecture (52 studies), reflecting their demonstrated effectiveness in image classification, detection, and segmentation tasks [11, 25]. Random Forest (15 studies) and Support Vector Machines (11 studies) remain widely used, particularly in studies involving multispectral or hyperspectral data where the feature space is structured and relatively low-dimensional. Architectures specifically designed for object detection such as R-CNN (4 studies) and for semantic segmentation such as U-Net (2 studies) also appear in the literature, reflecting the importance of spatially explicit analysis in agricultural monitoring. More recent architectures including Vision Transformers (ViT, 3 studies), attention-based models (4 studies), and generative adversarial networks (GAN, 2 studies) are present in smaller numbers, indicating active but still limited exploration of these approaches in agricultural contexts [24, 28, 27].

3.6 Machine Learning Tasks

Table 3 presents the distribution of machine learning tasks addressed in the reviewed studies. Classification is the most prevalent objective (38 studies), reflecting the dominance of categorical decision-making problems such as crop disease identification, growth stage assessment, and product quality evaluation. Detection tasks are similarly well represented (35 studies), highlighting the importance of spatially localised identification of targets including pests, lesions, and fruit instances within agricultural imagery. Prediction tasks account for 21 studies, encompassing applications such as yield forecasting, water stress quantification, and environmental variable estimation. Segmentation (10 studies) and regression (5 studies) appear with moderate frequency, supporting pixel-level crop mapping and continuous variable modelling, respectively. Tasks such as counting, data fusion and transfer learning, diagnostics, and recognition are each represented by one study, reflecting specialised objectives that have received comparatively limited attention.

The diversity of machine learning tasks observed across the literature underscores the multifaceted role of AI in agricultural image analysis, spanning descriptive, diagnostic, and predictive objectives. The dominance of classification and detection tasks is consistent with the prevalence of pest and disease detection as the leading agricultural application, while the growing presence of prediction and regression tasks reflects the field’s gradual expansion toward quantitative crop monitoring and decision-support applications.

Table 1: Distribution of AI model families across the reviewed studies. AI models were grouped into broader methodological families to facilitate a higher-level comparison of dominant computational approaches.

AI Model Family	Number of Studies
Deep Learning	54
Tree-based ML	16
Kernel-based ML	11
Other	9
Neural Networks (ML)	9
Regression	9
Boosting ML	7
Custom / Specialized	6
Dimensionality Reduction	5
Object Detection DL	5
Transformer-based DL	5
Distance-based ML	3
Generative Models	2
Reinforcement Learning	2

3.7 Comparative Performance Analysis Across Model Families

Table 4 synthesises the reported performance of the main AI model families across task types, based on metrics extracted from the 91 included studies. Because studies employ heterogeneous evaluation protocols and report different metrics (accuracy, F1-score, R^2 , mAP), comparisons across families and tasks must be interpreted with caution; nonetheless, several patterns emerge from the available evidence.

Deep learning approaches dominate in both frequency and reported performance for classification tasks, with accuracy and kappa values ranging from 0.82 to 0.99 (mean 0.94, $n = 29$), suggesting consistent high-performance outcomes when sufficient training data are available [11, 29, 26]. For detection tasks, deep learning models report mAP values spanning a wide range (0.05–1.00), reflecting heterogeneity in evaluation protocols and dataset complexity rather than inherent model variability. Classical machine learning methods — primarily Random Forest and SVM — achieve competitive accuracy in classification tasks (0.77–0.97, mean 0.87, $n = 5$), particularly in studies employing hyperspectral or thermal imagery where the feature space is structured and relatively low-dimensional [12, 30, 16].

Table 2: Distribution of individual AI models employed across the reviewed studies, considering unique studies per model.

AI Model	Number of Studies
CNN	52
Random Forest	15
SVM	11
ANN	9
XGBoost	6
Custom Model	6
PLS	5
Linear Regression	4
R-CNN	4
LSTM	3
PCA	3
Attention	3
K-means	3
KNN	3
ViT	3
GAN	2
U-Net	2
LDA	2
Reinforcement Learning	2
Transformer	2
GPR	2
AdaBoost	1
GRU	1
Decision Tree	1
DeepLabV3	1
BlendMask	1
SAM	1

For prediction tasks, regression and spectral models (PLS, GPR) report R^2 values of 0.66–0.98, comparable to deep learning approaches (0.82–0.83), suggesting that for continuous variable estimation with spectral inputs, classical approaches remain viable alternatives [30, 23, 19]. Hybrid and ensemble approaches show wider performance variability in prediction tasks (R^2 0.18–0.85), indicating that model combination does not automatically yield superior outcomes and that architecture design choices critically influence results [20, 22].

These patterns are consistent with the broader machine learning literature: deep learning excels when image-based feature extraction is required and training data are abundant, while classical methods retain competitive advantage for structured spectral data under data-constrained conditions. The continued co-existence of both paradigms in the reviewed literature therefore reflects rational methodological choices adapted to data availability and task requirements, rather than a simple linear transition from classical to deep learning approaches.

3.8 A Conceptual Framework for AI-Based Optical and Thermal Agricultural Image Analysis

Figure 4 presents a conceptual pipeline framework synthesising the analytical workflow common to the studies reviewed. The framework comprises seven sequential stages, from sensing modality selection through to deployment context. Annotations indicate where the reviewed literature is concentrated (Stages 3–5, shaded in the figure) and where critical methodological gaps persist (Stage 1–2 integration and Stage 7, highlighted with dashed borders). This framework is intended as a reference architecture to guide the design of future studies and to identify the structural gaps that currently limit the translation of research findings into operational precision-agriculture systems.

Table 3: Distribution of machine learning tasks addressed in the reviewed studies. The table summarises the primary analytical objectives pursued in the literature, independent of the specific AI models used.

Machine Learning Task	Number of Studies
Classification	38
Detection	35
Prediction	21
Segmentation	10
Regression	5
Estimation	2
Counting	1
Data Fusion / Transfer Learning	1
Diagnostics	1
Recognition	1

Table 4: Comparative performance of AI model families across task types, based on metrics reported in the 91 reviewed studies. Performance ranges reflect the minimum–maximum of parsed metric values; mean values are given in parentheses. Studies reporting metrics incompatible with cross-study normalisation (e.g. RMSE, MAE) are excluded from the range calculation. n = number of studies contributing to each cell.

Model Family	Task	n	Reported performance range (mean)
Deep Learning	Classification	29	Accuracy/Kappa: 0.82–0.99 (mean 0.94)
	Detection	22	mAP: 0.05–1.00 (mean 0.73)
	Prediction	3	R ² : 0.82–0.83 (mean 0.82)
	Segmentation	4	Accuracy: 0.86–0.96 (mean 0.92)
	Regression	1	R ² : 0.91
Classical ML (RF, SVM, KNN)	Classification	5	Accuracy/Kappa: 0.77–0.97 (mean 0.87)
	Detection	3	R ² : 0.64–0.79 (mean 0.71)
	Prediction	3	R ² : 0.69–0.88 (mean 0.78)
	Regression	3	R ² : 0.79
Regression / Spectral (PLS, GPR, LDA)	Prediction	3	R ² : 0.66–0.98 (mean 0.83)
	Detection	1	—
Hybrid / Ensemble	Classification	3	—
	Prediction	6	R ² : 0.18–0.85 (mean 0.64)

RF = Random Forest; SVM = Support Vector Machine; KNN = k -nearest neighbours; PLS = Partial Least Squares; GPR = Gaussian Process Regression; LDA = Linear Discriminant Analysis.
mAP = mean Average Precision. Performance values > 1 were normalised to $[0,1]$ prior to analysis.

The concentration of published work in the model selection, training, and evaluation stages (Stages 3–5) confirms that the field has reached a degree of methodological maturity in the core computational pipeline. By contrast, the integration of sensing modality selection with platform constraints (Stages 1–2) remains largely study-specific, with limited standardisation of preprocessing protocols or sensor calibration procedures. Stage 7 (deployment context) is the most underrepresented stage in the reviewed literature: the vast majority of studies are evaluated in offline or laboratory-like settings, with only a small proportion reporting results under real operational conditions. Bridging this gap between proof-of-concept demonstrations and field-deployable systems constitutes one of the most pressing priorities for the next generation of research in this domain.

Supplementary Material

The complete data extraction matrix for all 91 studies included in this review — comprising bibliographic information, sensor modality, acquisition platform, crop type, production scenario, AI model, computational task, target variable, evaluation metric, reported performance, and limitations — is

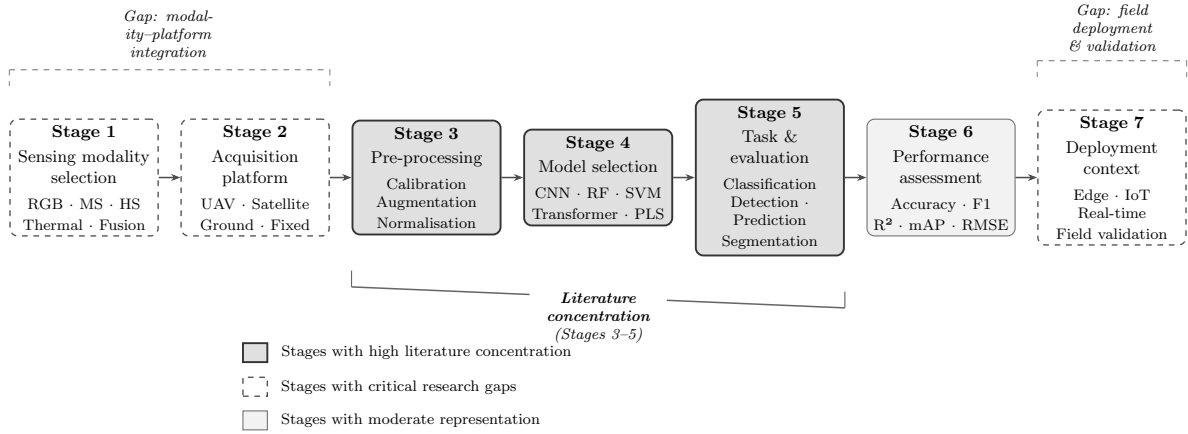


Figure 4: Conceptual pipeline framework for AI-based optical and thermal agricultural image analysis. The seven stages span the full analytical workflow from sensing modality selection to field deployment. Solid dark borders indicate stages where the reviewed literature is concentrated (Stages 3–5); dashed borders indicate stages with critical methodological gaps (Stages 1–2 integration and Stage 7). Abbreviations: MS = multispectral; HS = hyperspectral; UAV = unmanned aerial vehicle; CNN = convolutional neural network; RF = Random Forest; SVM = Support Vector Machine; PLS = Partial Least Squares; IoT = Internet of Things.

provided as Supplementary Material S1 (Excel file). This dataset supports full reproducibility of the frequency analyses presented in Sections 3.2–3.6 and the performance synthesis in Table 4.

4 Conclusions

This systematic review synthesised evidence from 91 peer-reviewed studies published between 2023 and 2025, characterising the technological configurations, algorithmic approaches, and application domains that define the current state of AI-based optical and thermal agricultural image analysis. The evidence reviewed supports several consolidated findings and points to a set of persistent structural gaps that constrain the translation of research advances into operational precision-agriculture systems.

RGB sensors and UAV-based acquisition platforms emerge as the dominant technological configuration, appearing in 30 and 32 studies respectively. This prevalence reflects their relative accessibility, low operational cost, and broad compatibility with existing AI frameworks. Multisensor configurations — combining RGB with thermal or spectral modalities — represent the second most common sensing approach (26 studies), confirming a growing recognition that complementary sensor fusion yields more robust and context-sensitive diagnostic outputs than single-modality pipelines. Thermal imaging, despite its specific utility for physiological stress detection and irrigation management, remains a secondary modality in absolute terms, appearing as the primary sensor in 13 studies while contributing as a fused component in a further 13.

Deep learning constitutes the overwhelmingly dominant methodological paradigm, with 54 studies employing deep learning architectures and CNNs present in 52 of the 91 included works. The comparative performance analysis presented in Table 4 reveals that deep learning achieves consistently high classification accuracy (0.82–0.99, mean 0.94) when training data are sufficient. However, classical machine learning methods — notably Random Forest and SVM — retain competitive performance for classification tasks involving structured spectral data (0.77–0.97, mean 0.87), and regression and spectral models (PLS, GPR) yield R^2 values comparable to deep learning for continuous variable prediction (0.66–0.98). The co-existence of both paradigms in the literature therefore reflects rational methodological adaptation to data availability and task requirements, rather than a simple linear methodological transition.

From an application perspective, pest and disease detection, irrigation and water status monitoring, and crop maturity and phenology assessment collectively account for the majority of the reviewed

studies. The distribution of research across crop types is markedly uneven: cereal crops dominate the literature, while legumes, vineyards, tropical fruit crops, and agroforestry systems remain systematically underrepresented. The conceptual pipeline framework proposed in Figure 4 further reveals that the reviewed literature is concentrated in the model selection, training, and evaluation stages (Stages 3–5), while the integration of sensing modality selection with platform constraints (Stages 1–2) and the validation of systems under real operational deployment conditions (Stage 7) represent the most critical underexplored areas.

Several limitations of the present review should be acknowledged. The search was restricted to ScienceDirect and IEEE Xplore; the inclusion of additional databases such as Scopus, Web of Science, or PubMed may yield further relevant studies, particularly from plant science and agronomy journals. The review covers a three-year window (2023–2025), which, while suited to capturing recent methodological trends, limits historical perspective. Performance comparisons across model families (Table 4) are based on heterogeneous evaluation protocols and should be interpreted as indicative rather than definitive benchmarks.

Taken together, these findings confirm that AI-based optical and thermal agricultural image analysis has reached a meaningful degree of methodological maturity in its core computational pipeline. Realising the full potential of this maturity for operational agricultural systems will require coordinated progress on four fronts: the construction of standardised, open benchmark datasets spanning multiple sensing modalities and crop categories; the development of multimodal sensor fusion architectures capable of jointly exploiting RGB, thermal, and spectral information; the systematic evaluation of AI models under real field deployment conditions; and the expansion of research coverage to underrepresented crops and geographic regions. Addressing these priorities will be essential to translate the considerable research advances documented in this review into scalable, reliable precision-agriculture tools accessible across diverse agricultural contexts.

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